MTH 201 Chapter 3: Infinite Series

3.6 Alternating Series

So, far we have been focusing on series which ultimately have all positive terms. In this section we will consider a special kind of series that has negative terms.

Definition: An alternating series is a series whose terms are alternately positive and negative.

Example: The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$ is an alternating series.

The series $\sum_{n=1}^{\infty} \frac{(-1)^{n} n}{n+1} = -\frac{1}{2} + \frac{2}{3} - \frac{3}{4} + \frac{4}{5} - \cdots$ is an alternating series.

Note: We can write the nth term of an alternating series in the form $(-1)^{n-1} a_n$ or $(-1)^n a_n$.

Example 1: Consider the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.

a) Using the notation from class what does a_n equal in this case?

It is $a_n = \frac{1}{n}$.

b) TRUE or FALSE: $a_{n+1} \le a_n$ for all n and $\lim_{n\to\infty} a_n = 0$.

Based upon our knowledge of the harmonic series, this is TRUE.

c) Compute the first, second, third, fourth, fifth, and sixth partial sum of this series.
We see that:

$$s_1 = 1$$
$$s_2 = 1 - \frac{1}{2} = \frac{1}{2}$$
$$s_3 = 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$$
$$s_4 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} = \frac{7}{12}$$
$$s_5 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} = \frac{47}{60}$$
$$s_6 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} = \frac{37}{60}.$$

d) TRUE or FALSE: The even partial sums for this series form an increasing sequence.

We note that for $k \geq 1$, $s_{2k} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2k-1} - a_{2k})$. Since $a_{n+1} \leq a_n$ for all n we know that each term in parentheses is positive. Thus, we add a positive number to get the next term in the sequence of even partial sums of the series. This means that the sequence is increasing and the answer is TRUE.

e) TRUE or FALSE: The sequence of even partial sums for this series is bounded above by 1.

This is TRUE. The reason is because we start with 1 as our first term of the series and then alternate between adding and subtracting positive numbers which decrease in size (see the textbook for a nice illustration of this). Mathematically we can see this by noting that for $k \geq 1$:

$s_{2k} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots - (a_{2k-2} - a_{2k-1}) - a_{2k} \leq a_1 = 1$

f) Do you think this series is convergent or divergent? Explain.

By the Monotonic Sequence Theorem we note that the sequence of even partial sums must converge (by parts (d) and (e)). In particular suppose $\lim\limits_{k \to \infty} s_{2k} = s$. Then,

$\lim\limits_{k \to \infty} s_{2k+1} = \lim\limits_{k \to \infty} (s_{2k} + a_{2k+1}) = \lim\limits_{k \to \infty} \left(s_{2k} + \dfrac{1}{2k+1} \right) = s$. Since the sequence of even partial sums for the series and the sequence of odd partial sums for the series converge to the same limit, we conclude that the series is convergent.

Note: This example uses many of the ideas in the proof of the Alternating Series Test.

Theorem (Alternating Series Test): If the alternating series $\sum\limits_{n=1}^{\infty} (-1)^{n-1} a_n$ (where $a_n > 0$) satisfies the following two conditions:

1. $a_{n+1} \leq a_n$ for all n
2. $\lim\limits_{n \to \infty} a_n = 0$

then the series is convergent.

Proof: See the textbook..

Note: The exponent on the negative 1 can be replaced with n plus or minus a natural number and the theorem will still hold.

Note: The theorem will still hold if the index starts at a natural number other than 1.

Example: The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is convergent by the Alternating Series Test since from the last example we know that for this series $a_n = \frac{1}{n}$, $a_{n+1} \le a_n$ for all n, and $\lim_{n\to\infty} a_n = 0$.

Example 2: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^{n+1}(n+3)}{n(n+1)} \right)$ converges or diverges.

We immediately recognize that for this alternating series $a_n = \frac{n+3}{n(n+1)}$. We want to know if this sequence is decreasing so we will investigate the ratio of consecutive terms.

$\frac{a_{n+1}}{a_n} = \frac{n+4}{(n+1)(n+2)} \cdot \frac{n(n+1)}{n+3} = \frac{n(n+4)}{(n+2)(n+3)} = \frac{n^2+4n}{n^2+5n+6} = \frac{n^2+4n}{\left(n^2+4n\right)+\left(n+6\right)} < 1$. Therefore we see that $a_{n+1} \le a_n$ for all n.

Now, we also calculate that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n+3}{n(n+1)} \cdot \frac{1/n^2}{1/n^2} = \lim_{n\to\infty} \frac{\frac{1}{n}+\frac{3}{n^2}}{1+\frac{1}{n}} = \frac{0}{1} = 0$.

So, by the Alternating series test, the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^{n+1}(n+3)}{n(n+1)} \right)$ converges.

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Example 3: Determine (with justification) whether the series $\sum_{n=1}^{\infty} \left(\frac{(-1)^{n}(3n-1)}{2n+1} \right)$ converges or diverges.

We immediately recognize that for this alternating series $a_n = \frac{3n-1}{2n+1}$. We calculate that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3n-1}{2n+1} \cdot \frac{1/n}{1/n} = \lim_{n\to\infty} \frac{3-1/n}{2+1/n} = \frac{3}{2}$. So, the Alternating Series Test does not say anything about this series.

Since the terms of this series do not get arbitrarily close to zero we suspect that this series diverges. Indeed, we see that if $a_n = \frac{(-1)^n(3n-1)}{2n+1}$, the odd terms get arbitrarily close to $\frac{3}{2}$ and the even terms get arbitrarily close to $\frac{-3}{2}$. Thus, $\lim_{n\to\infty} a_n$ does not exist and the series diverges by the Test for Divergence.

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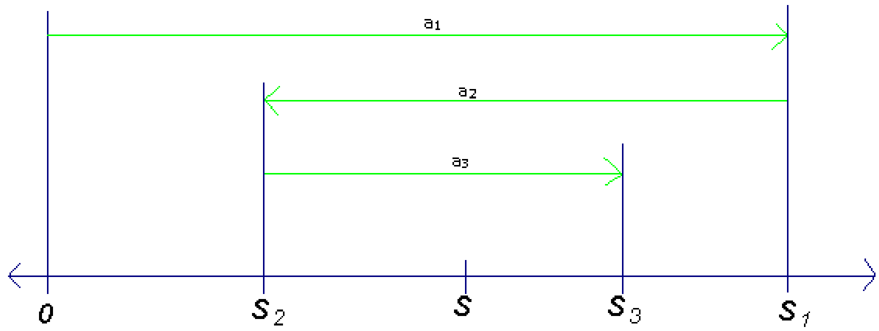
Theorem: If $s=\sum_{n=1}^{\infty}(-1)^{n-1}a_n$ (where $a_n>0$) is the sum of an alternating series that satisfies $a_{n+1}\le a_n$ for all n and $\lim_{n\to\infty}a_n=0$, then $|R_n|=|s-s_n|\le a_{n+1}$.

We will omit the proof of this theorem.

Note: Intuitively the above theorem works since the partial sums of an alternating series satisfying the above conditions oscillate about the actual sum. This idea is captured in the picture below.



Example 4: Consider the series $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}$.

a) Find the seventh partial sum of this series.

It is: $\sum_{n=1}^{7}\frac{(-1)^{n-1}}{n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\frac{1}{7}=\frac{319}{420}\approx .7595$.

b) We know that this series converges. What is an upper bound on how far off the number that you calculated in (a) is from the actual sum of the series.

If s is the sum of the series we know from the above theorem that $|s-s_7|\le\frac{1}{8}=.125$. Thus, we know that $\frac{319}{420}$ is within an eighth of the actual sum.

c) What is the first partial sum of this series that we can guarantee is within one thousandth of the actual sum (using the theorem)?

The answer to this question is the 999$^{\text{th}}$ partial sum since by the theorem we know that $|s-s_{999}|\le b_{1000}=\frac{1}{1000}$.

Note: Later on in this course we will be able to prove that $\sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n}=\ln(2)$.

Definition: A series $\sum a_n$ is called <u>absolutely convergent</u> if the series of absolute values $\sum |a_n|$ is convergent.

A series $\sum a_n$ is called <u>absolutely divergent</u> if the series of absolute values $\sum |a_n|$ is divergent.

A series $\sum a_n$ is called <u>conditionally convergent</u> if it is convergent but not absolutely convergent (absolutely divergent)

<u>Example 5:</u> State (with justification) whether each of the following series is absolutely convergent, conditionally convergent, or divergent.

a) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$

This series is not absolutely convergent (absolutely divergent) since $\displaystyle\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$ and we know the harmonic series is divergent. It is however conditionally convergent since we know that $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ is convergent by the Alternating Series Test.

b) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$

This series is absolutely convergent since we know that $\displaystyle\sum_{n=1}^{\infty} \left| \frac{(-1)^{n-1}}{n^2} \right| = \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the *p*-series test.

Theorem: If a series $\sum a_n$ is absolutely convergent, then it is convergent.

Proof: We first note that $0 \le a_n + |a_n| \le 2|a_n|$. Since $\sum a_n$ is absolutely convergent, we know that $\sum |a_n|$ is convergent. By the limit laws for series this implies that $\sum 2|a_n|$ is convergent. Then, by the Comparison Test, we know that $\sum (a_n + |a_n|)$ is convergent. Finally, by the limit laws for series we know $\sum (a_n + |a_n|) - \sum |a_n| = \sum a_n$ is convergent.

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Example 6: Show that the following series converges: $1 - \frac{1}{2} - \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} - \frac{1}{2^5} - \frac{1}{2^6} + \ldots$

We should first note that this series is not alternating, so the alternating series test does not apply. However, the series of absolute values is the geometric series $\sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^{n-1} = 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \frac{1}{2^5} + \frac{1}{2^6} + \ldots$, which is convergent since $|r| < 1$. Therefore, since the given series is absolutely convergent it is convergent.

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